

2.7 Specialist techniques, tools, equipment and processes that can be used to shape, fabricate, construct and assemble a high-quality ferrous and/or non-ferrous metal prototype	Application, advantages and disadvantages of the following specialist techniques when manufacturing products, in order to be able to discriminate between them and select appropriately for use.	
	2.7.1	Tools and equipment: a hand tools b machinery c digital design and manufacture.
	2.7.2	Shaping: a filing b cutting/shearing c drilling d turning e milling f bending g abrading/grinding h casting i deforming and reforming.
	2.7.3	Fabricating/constructing: a welding b brazing c soldering d stamping e punching f riveting snap and pop g sheet metalwork h wastage i addition.

METALS

Key idea	What students need to learn:
	2.7.4 Assembling: - a tapping/threading - b fastening – use of nuts, bolts and washers - c machine screws - d use of adhesives – contact adhesive, epoxy resin.